

Swiss Water Monitoring Agencies

Sensor Networks, Data Formats & Open Data Status

41 monitoring networks · Federal, cantonal & research institutions

August 2026

This document catalogues 41 Swiss water monitoring networks maintained by federal agencies (BAFU/FOEN, MeteoSwiss, swisstopo), cantonal environmental offices, research institutions (Eawag, WSL), and community aggregators. For each network it records the water body type, parameters measured, station count, measurement frequency, data format, open-data status and API / access method — drawing exclusively on sources fetched in August 2026. Agency-level context and known gaps are discussed in the sections following the master table.

Coverage at a glance

- Federal networks (BAFU/FOEN, NAWA, NAQUA, MeteoSwiss, swisstopo): 12 networks
- Cantonal networks (Zürich, Bern, Vaud, Ticino, Geneva, Aargau, Luzern, Thurgau, Basel): 17 networks
- Research / Eawag networks (UWO, Datalakes, LÉXPLORE, Aquascope, wastewater, WSL): 9 networks
- Aggregators & portals (Existenz.ch, opendata.swiss, WVZ, SVGW): 3 networks

Agency	Network	Water Body	Parameters	Stations	Frequency	Format	Open Data	API / Access
BAFU/FOEN, Abteilung Hydrologie (Bundesamt für Umwelt)	Basismessnetz Oberflächengewässer (BAFU)	Rivers + lakes (geocat)	Water level; discharge; also water temperature, chemical composition, turbidity at stations (BAFU)	~260 stations, discharge determined at 200 sites (BAFU); "a total of 260 surface water gauging	Continuous with remote transmission; 10-minute updates for published data on LINDAS (hydrodaten); weekly station check by local	Station directories as XLSX (BAFU); station geodata as ESRI SHP and FileGDB (geocat)	Yes — "Opendata OPEN: Freie Nutzung" / "Open use" for the station dataset (geocat)	STAC browser 'data.geo.admin.ch', ESRI REST 'api3.geo.admin.ch/rest/service WMS/WMTS (geocat); live values via LINDAS Linked Data Service and the "Meine Pegel" app (hydrodaten)
BAFU/FOEN	Messnetz Wassertemperatur (Fließgewässer) (BAFU)	Rivers (opendata.sw	Water temperature (BAFU)	~80 stations (BAFU); "rund 80 Messstationen (hydrodaten)	Continuous; dataset update interval "kontinuierlich" (opendata.swiss)	Metadata/stat data via API (JSON) and XML download (opendata.sw	Partial — dataset published on opendata.swi but no licence named (opendata.sw	Landing page 'hydrodaten.admin.ch'; API (JSON) / XML (opendata.swiss); temperatures also republished 10-min via api.existenz.ch (Existenz)
BAFU/FOEN	Pilotprojekt Seewassertemperatur (BAFU)	Lakes, small ponds (BAFU)	Water temperature incl. depth profile at deepest point (BAFU)	3 pilot stations (Murtensee, Hallwilersee, Ägerisee) plus 8 further small lakes (BAFU)	Continuous at several depths (BAFU)	n.a.	n.a.	n.a.
BAFU/FOEN	Hydrologische Untersuchungsgebiete der Schweiz (HUG) (BAFU)	Rivers in near-natural catchments (BAFU)	Natural runoff depths; monthly and annual catchment precipitation; annual catchment evaporation; catchment characteristics (BAFU)	~40 study catchments (BAFU)	Monthly/annual aggregates; network operated since 1957 (BAFU)	n.a.	n.a.	Station dataset "Gauging stations for hydrological study areas" listed on opendata.swiss (opendata.swiss search hit)
BAFU/FOEN (with cantons)	NAWA TREND (Basismessnetz Oberflächengewässerqua (BAFU)	Rivers and streams (BAFU EN)	Nutrients; micropollutants; biological indicators; chemical-physical parameters (BAFU)	~140 monitoring sites (BAFU EN)	Long-term continuous monitoring; chem-phys since 2011, biology first 2012, micropollutants at selected sites since 2018 (BAFU)	Station list published as XLSX ("Messstellen ... NAWA TREND 2026") (BAFU EN)	Partial — station lists, substance lists and BAUFU parameter nomenclature (v. 11/2025) published; no	Download of documents/station lists from the BAUFU NAWA page; no API stated (BAFU)
BAFU + Eawag + WSL	NAWA FRACHT (formerly NADUF) (BAFU)	Rivers (large, medium and, since the 2000s, smaller watercourses (BAFU)	Continuous: water level (→ discharge), water temperature; at most stations pH, electrical conductivity, dissolved oxygen via measuring electrodes. 14-day composite samples: hardness, alkalinity, Ca, Mg, K, Na, Cl, SO ₄ , H ₄ SiO ₄ , NO ₃ , PO ₄ , P _{tot} , DOC, TOC, N _{tot} (since 1987); heavy metals Zn, Cu, Pb, Cd, Hg, Cr, Ni at some stations (BAFU)	Total not stated ("selected stations of the BAUFU network"); plus 3 WSL Alptal stations	Sensors continuous; chemistry on 14-day composite samples (BAFU); weekly water-quality data for the three Alptal torrents since 2003	Programme documentation as PDF ("NAWA FRACHT: Messnetzbetr 2025") (BAFU)	Partial — Alptal weekly water-quality data downloadable from the NADUF website (WSL); no	Direct download for Alptal NADUF data (WSL)

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BAFU/FOEN	NAWA SPEZ (problem-related special monitoring) (BAFU)	Rivers, incl. small streams (BAFU)	Pesticides; biological state of small streams (BAFU)	Selected sites; number not stated (BAFU EN)	Campaign-based / ad hoc: pesticide campaigns 2012, 2015, 2017, 2023; small-stream biology 2018 (BAFU)	n.a.	n.a.	n.a.
BAFU/FOEN	NAQUA — Modul QUANT (BAFU)	Groundwater + springs (geocat)	Groundwater levels; spring discharge; water temperature; electrical conductivity; partly turbidity (BAFU)	91 monitoring sites; half federal / half cantons or water utilities; 30 also in TREND, 17 in SPEZ (BAFU); geodata	Continuous recording of temperature, conductivity and partly turbidity; a few cantonal sites still read out periodically (delayed)	Station geodata as GeoJSON; WMS layer 'ch.bafu.hydr (geocat)	Partial — station geodata published via geo.admin.ch measurement licence not stated (geocat)	GeoJSON + OGC WMS + map.geo.admin.ch preview (geocat); station pages on hydrodaten.admin.ch (hydrodaten Wila)
BAFU/FOEN	NAQUA — Modul TREND (BAFU)	Groundwater, springs (BAFU)	Various organic and inorganic compounds and elements incl. calcium, magnesium, bicarbonate; water levels; spring discharge (BAFU)	50 detailed monitoring sites (BAFU)	n.a. (module centrally operated since 1997, fully operational since 2005) (BAFU)	n.a.	No — no data licence or download stated (BAFU)	n.a.; results published as PDF report (BAFU)
BAFU/FOEN (with cantons)	NAQUA — Modul SPEZ (BAFU)	Groundwater (BAFU)	Nitrate; pesticide residues; pharmaceutical residues; other organic trace substances; halogenated hydrocarbons (BAFU)	~500 ("knapp 500") sites across Switzerland (BAFU)	n.a. (operated with the cantons since 2002) (BAFU)	n.a.	No — none stated (BAFU)	n.a.
BAFU/FOEN (with MeteoSwiss sites)	NAQUA — Modul ISOT (BAFU)	Precipitation + rivers (BAFU)	Stable water isotopes oxygen-18 and deuterium; at co-located MeteoSwiss sites also air temperature and vapour pressure (BAFU)	22 sites: 13 precipitation + 9 river sites (BAFU)	n.a.	n.a.	No — none stated (BAFU)	n.a.
BAFU/FOEN (portal)	hydrodaten.admin.ch — Hydrologische Daten und Vorhersagen (hydrodaten)	Rivers, lakes, groundwater (hydrodaten)	Water level, discharge, water temperature, flood danger levels; water quality (hydrodaten), (BAFU)	Aggregates the BAFU networks; portal itself states no station count (hydrodaten)	Real-time raw data; LINDAS updated every 10 minutes (hydrodaten)	Station-status dataset offered as API (JSON) / XML on opendata.swi (opendata.swi "Hydrodaten Open Data	Partial — data are published as open data and republished via LINDAS, but no licence name stated on the	LINDAS Linked Data Service (contact abfragezentrale@bafu.admin.ch "Meine Pegel" app, station web pages (hydrodaten)

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MeteoSwiss (Bundesamt für Meteorologie und Klimatologie)	SwissMetNet automatic measurement network (MeteoSwiss opendata docs)	Atmosphere — not water bodies; MeteoSwiss states monitoring of lakes and rivers is not its remit	Air temperature, precipitation, wind, sunshine, humidity, radiation, pressure (MeteoSwiss)	~160 SwissMetNet stations + ~100 automatic precipitation stations + 3 tower stations	Every 10 minutes (MeteoSwiss)	Open-data files, republished as JSON by api.existenz.ch (Existenz)	Yes — published as OGD by MeteoSwiss (MeteoSwiss) licence name not stated on the fetched page	opendatadocs.meteoswiss.ch download service; third-party 10-min JSON API at api.existenz.ch (Existenz)
swisstopo (Bundesamt für Landestopografie)	swissTLM3D Gewässernetz / hydrographic network (opendata.swiss)	Rivers + lake outlines (Switzerland + Liechtenstein (opendata.swiss))	None — spatial reference data, no measured parameters or sensors (opendata.swiss)	n.a. (no stations) (opendata.swiss)	Dataset updated annually (opendata.swiss)	API (JSON) / XML metadata access (opendata.swiss)	Partial — published on opendata.swiss licence not stated on the page (opendata.swiss)	opendata.swiss / swisstopo landing page; geo.admin.ch services (opendata.swiss)
AUE Basel-Stadt + BAFU + LUBW Baden-Württemberg	Rheinüberwachungsstation Weil am Rhein (RUS) (Kanton Basel-Stadt)	River (Rhine, km 171.37) (data.bs.ch)	680 pollutants monitored regularly, 420 of them daily (Kanton Basel-Stadt); plus continuously measured sensor parameters (data.bs.ch)	1 binational station (network total not stated) (Kanton Basel-Stadt)	Continuous, averaged and published every 15 minutes as raw data; annually plausibilised hourly values (data.bs.ch)	Open-data portal dataset '100046' with API console and downloads (data.bs.ch)	Yes — results publicly accessible via the Basel-Stadt open-data portal (Kanton Basel-Stadt);	data.bs.ch dataset pages + API console; further datasets e.g. Birs ('100236') and Rhein level ('100089') (data.bs.ch Birs)
AWEL Kanton Zürich (Amt für Abfall, Wasser, Energie und Luft), Sektion Geoinformation und Hydrometrie	Hydrometrische Stationen (kantonales Messnetz) (Kanton Zürich)	Rivers + lakes (Kanton Zürich)	Water level; discharge (derived from level via station-specific rating curve, checked by calibration measurements) (Kanton Zürich)	Over 60 stations (Kanton Zürich)	Continuous recording; data online in near real time or at latest after one day; curves for last 24 h / 7 d / 31 d (Kanton Zürich)	n.a. — data delivered on request; metadata viewable at maps.zh.ch (Kanton Zürich)	Partial — declared Open Government Data (OGD) but no licence named and delivery via	Web viewer + order form; metadata layer "Hydrometrische Stationen" on maps.zh.ch; requests normally free, CHF 140.20/h for very large orders; attribution "AWEL, Kanton Zürich" (Kanton Zürich)
AWEL Kanton Zürich, Gewässerschutzlabor	Gewässermonitoring / Untersuchungsprogramm Seen und Fliessgewässer 2022–2025 (Kanton Zürich)	Lakes, rivers, groundwater + spring water (Kanton Zürich)	Water quality assessed with the federal Modul-Stufen-Konzept; individual parameters not listed on the page; micropollutants and ecomorphology covered (Kanton Zürich)	11 lakes, 150 river sites, ~100 groundwater/abstractions (Kanton Zürich); 15 main river stations	Continuous automatic sampling at the 15 main stations; periodic at the other 150 sites; each catchment once per 4-year period; annual evaluation, main	PDF reports (programme 35 pp., methods 31 pp.) (Kanton Zürich)	No — reports/maps only, no API and no licence stated (Kanton Zürich)	Web maps with per-station reports (Kanton Zürich)
AWEL Kanton Zürich, Abteilung Gewässerschutz	Grundwasserstände (kantonales Grundwasserbeobachtungsnetz) (Kanton Zürich)	Groundwater (Kanton Zürich)	Groundwater level; mean water level maps with isohypses (Kanton Zürich)	n.a. (Kanton Zürich)	Continuous ("kontinuierlich") (Kanton Zürich)	n.a.	n.a.	Web map with newest values, geological borehole profiles (Kanton Zürich)

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AWEL Kanton Zürich, Sektion Grundwasser und Wasserversorgung	Grundwasserqualität (kantonale Umweltbeobachtung + NAQUA) (Kanton Zürich)	Groundwater, spring water, drinking-water abstractions (Kanton Zürich)	Nitrate, hardness, conductivity, temperature, oxygen, pH, carbonic acid, Cl^- , NO_2^- , NH_4^+ , SO_4 , PO_4 ; PFAS; chlorothalonil metabolites R417888/R471811; organic trace substances (Kanton Zürich)	Selected sites, mostly groundwater abstractions; number not stated (Kanton Zürich)	n.a.; chlorothalonil metabolites monitored since 2019, published since 27.03.2023 (Kanton Zürich)	Web maps ("Züri Trinkwasser Map") (Kanton Zürich)	Partial — results shown on public maps, no API/licence stated (Kanton Zürich)	Four regional map viewers (NW/NE/SW/SE) (Kanton Zürich)
AWA Kanton Bern (Amt für Wasser und Abfall)	Hydrologische Messnetze des AWA (Oberflächengewässer, Grundwasser, Niederschlag, Pegelmessnetz) (Kanton Bern)	Rivers, lakes, groundwater, precipitation (Kanton Bern)	Discharge, water level, water temperature; groundwater level and groundwater temperature; precipitation and air temperature (Kanton Bern)	~50 discharge stations, 90 groundwater stations, 120 water-level gauges, 15 precipitation stations	n.a.; monthly regional groundwater evaluation published (Kanton Bern)	Station lists/maps via cantonal geoportal; bulletins and yearbooks (Kanton Bern)	Partial — data published online with federal data; delivery of measurement series via "Datenbezug ordering"	Geoportal maps + order channel; flood portal for current situation (Kanton Bern)
DGE Canton de Vaud (Direction générale de l'environnement)	Suivi de la qualité des rivières / stratégie de surveillance des eaux (info.vd.ch)	Rivers (info.vd.ch)	Aquatic macroinvertebrates (biological quality); nutrients (carbon, nitrogen, phosphorus); micropollutants (pesticides, pharmaceuticals, industrial products) (info.vd.ch)	204 sites on more than 50 rivers (info.vd.ch)	n.a.; monitoring run since the 1990s (Etat de Vaud)	Web platform; reports as PDF (Etat de Vaud)	Partial — results published on www.vhv-qu no licence and no API stated (info.vd.ch)	Web platform vhw-qualite.ch; results also on the "Géoportail de la veille hydrologique vaudoise" (Etat de Vaud)
Canton Ticino, Ufficio dei corsi d'acqua (UCA)	Rete di misura cantonale (idrometria) (Repubblica e Cantone Ticino)	Rivers + lakes (Lago Maggiore, Lago di Lugano) (Ticino)	Discharge (m^3/s), water temperature, lake level, precipitation (OASI), (Ticino)	24 hydrometric + 31 meteorologic cantonal stations (Ticino); OASI portal lists 39 river	Half-hourly, daily, monthly and annual values available; quarterly bulletin, annual yearbook (OASI), (Ticino)	Portal download function ("Selezione avanzata e scaricamento dati"); yearbooks as PDF (OASI)	Partial — free portal download, no licence named, no API (OASI)	OASI web portal (station + parameter + interval selection, station comparison, data download) (OASI)
Canton Ticino, Ufficio protezione acque e approvvigionamento idrico (UPAAI)	Monitoraggio corsi d'acqua — Sistema modulare a livelli (Ticino)	Rivers/stream (Ticino)	Nitrogen compounds; total phosphorus; "external aspects" module (sludge, turbidity, colour, foam, odour, iron sulphide, clogging, litter, heterotrophic vegetation); ecomorphology (Ticino)	986 watercourses analysed = 17 % of the ~5,800 km cantonal river network; station count not stated	Campaign-based: 2003–2005 first campaigns, most recent 2021–2022 (Ticino)	Field check-lists; no data formats stated (Ticino)	No — no licence, API or download stated (Ticino)	n.a. (Ticino)

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Canton de Genève (hydro-meteorological monitoring)	Veille hydrologique / portail de suivi hydro-météorologique (Etat de Genève)	Lake + rivers (Etat de Genève)	Precipitation, water level, discharge, lake level, temperature, low-flow indicators (Etat de Genève)	37 stations: 12 rain-gauge + 25 limnimetric (Etat de Genève)	Real-time monitoring; monthly bulletins; weekly low-flow follow-up in summer; annual validation of previous year (Etat de Genève)	n.a. (Etat de Genève)	Partial — validated historical data can be extracted automatically per station; current-year validated data	Web portal with automatic historical extraction; station geodata on SITG (SITG)
Kanton Aargau (Abteilung Landschaft und Gewässer, Fachbereich Hydrometrie)	Hydrometrisches Messnetz Aargau (Kanton Aargau)	Rivers (Kanton Aargau)	Water level (redundant), water temperature, calibration measurements (Kanton Aargau)	n.a. — station overview per station page (e.g. Furtbach-Wü FG_0548) (Kanton Aargau station)	Transmitted every 10 minutes; time series downloadable at 10-minute resolution (Kanton Aargau)	"Standardisie Formate" for download; format names not stated (Kanton Aargau)	Yes — publicly available "für den freien Gebrauch"; no licence name stated (Kanton Aargau)	Station web app with data download (ag.ch/app/hydrometrie/station) (Kanton Aargau station)
Kanton Aargau (groundwater)	Grundwasserdaten / gemeinsames Messnetz AG-LU (Kanton Aargau)	Groundwater (+ precipitation, surface water) (Kanton Aargau)	Groundwater level, discharge, water level, precipitation; biological and chemical water-quality indicators (Kanton Aargau)	Over 50 abstraction sites available as online stations (Kanton Aargau)	Daily resolution, updated daily (Kanton Aargau)	Excel download for the last ten years of groundwater levels; hydrological yearbook electronic	Partial — free download of Excel series and interactive maps; no licence stated (Kanton Aargau)	Interactive maps + Excel download (Kanton Aargau)
Kanton Luzern, Dienststelle Umwelt und Energie (uwe) with vif	Wassermessstellen / Hydrometrie Luzern (geocat)	Rivers, lakes, groundwater, springs (+ precipitation) (geocat)	Water level and discharge of rivers; lake level; groundwater level; spring discharge; precipitation (geocat)	14 river level/dischar sites; ~40 groundwater- and spring-discha sites; 10 cantonal precipitation	n.a.; records partly go back to 1970 (geocat)	Station geodata as ESRI GDB, SHP, OGC GeoPackage (geocat); other formats on request, chargeable	Partial — station geodata on the cantonal geoportal; measurement data via order form, chargeable	Geoportal geo.lu.ch + "Bestellung Hydrologiedaten" form (Kanton Luzern)
Kanton Thurgau, Amt für Umwelt	Hydrologisches Messnetz Thurgau (data.tg.ch)	Rivers/groun (types not stated in dataset) (data.tg.ch)	Water level (m a.s.l.) and water temperature (°C) as separate datasets (data.tg.ch temperature)	n.a. — "verschieden Messstationen one object ID per parameter (data.tg.ch)	Live datasets hold the last 365 days; finer granularity in full downloads (data.tg.ch)	Zipped CSV for full datasets; JSON schema documented (data.tg.ch)	Yes — published on the cantonal OGD portal (also mirrored on opendata.swi licence not named on the	Opendatasoft portal (dataset pages + API console), station map via ThurGIS (data.tg.ch)

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Eawag (Swiss Federal Institute of Aquatic Science and Technology) + ETH Zurich	Urban Water Observatory (UWO), Fehraltorf (Eawag)	Wastewater/s rivers (Luppmen/Ke groundwater (ESSD))	Rainfall, flow, water level, conductivity, temperature (incl. in-sewer atmosphere), proxy pollution-load sensors, pesticide grab/passive samples; planned pH, conductivity, turbidity in water quality (Co-UDlabs), (Eawag)	124 sensors in the published dataset, of which 89 transmit via LoRa, with 3 LoRaWAN gateways	1-5 minute resolution (Co-UDlabs)	Eawag "Datapool" data warehouse with data-access packages for Julia, Python and	Yes — "unique open-access data set", data and metadata publicly available (Eawag);	Datapool database + DAP libraries + web dashboard; facility access "partially remote", 20 days per project (ESSD), (Co-UDlabs)
Eawag (+ partner institutions)	Datalakes platform / "Lake Monitoring on Swiss Lakes (2019-ongoing)" (Datalakes)	Lakes / surface waters (Datalakes)	In-situ physical, biological and chemical measurements; lake water temperature as the core national long-term target (Eawag), (Datalakes)	n.a. (Datalakes)	"High-frequency" observations; no interval stated (Datalakes)	n.a. — portal offers dataset downloads; formats not stated (Datalakes data portal)	Yes — "open-access sensor-to-fro platform", built on open science and FAIR principles; licence name	Interactive map viewer, data portal with spatial/temporal filters, and an API (Datalakes), (Datalakes data portal)
Eawag + EPFL + UNIL + UNIGE	LéXPLORE floating platform, Lake Geneva (Eawag)	Lake (Lake Geneva, near Pully, 110 m depth) (Eawag)	Lake physico-biological processes and water-atmosphere interactions; skin temperature and sky temperature series published on Datalakes (Eawag), (Datalakes data portal)	1 platform (10-year deployment) (Eawag)	"Unparalleled frequency", high temporal/spatial resolution; live skin-temperature feed (Eawag), (Datalakes data portal)	Via Datalakes datasets (Datalakes data portal)	Partial — core datasets published through Datalakes; licence n.a. (Datalakes data portal)	Datalakes portal/API (Datalakes data portal)
Eawag	Aquascope, Greifensee (Eawag)	Lake (Greifensee) (Eawag)	Images of planktonic microbes (Datalakes)	1 instrument site (Eawag)	High-frequency imaging, photographing since spring 2018; live images (Eawag)	Images (Eawag)	Partial — live images viewable on the Aquascope platform (Eawag)	Aquascope web platform (Eawag)
Eawag + BAG (Federal Office of Public Health)	Nationales Abwassermonitoring (respiratory viruses; formerly AbwasSARS-CoV-2) (BAG)	Wastewater (WWTP influent / raw sewage) (Eawag)	SARS-CoV-2, influenza A, influenza B, RSV; absolute viral load (gene copies per day per 100,000 inhabitants), 7-day moving average, relative viral load, variants/mutations by sequencing; WWTP flow volume for dilution (BAG)	10 WWTPs in the regular programme since January 2025, covering 1.97 M people (~a quarter of the population),	Daily raw-sewage sampling in the original programme; series updated one to two times per week (Eawag)	Dashboard + open datasets on opendata.swi (opendata.swi)	Yes — datasets published on opendata.swi (opendata.swi) licence n.a.	BAG infectious-disease dashboard; variant data on covSPECTRUM since 2026 (BAG)

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Eawag	DroMedArio (wastewater-based epidemiology) (Eawag)	Wastewater (Eawag)	Drugs, medicinal products, alcohol and tobacco residues (Eawag)	n.a. (Eawag)	n.a. ("monitoring over time") (Eawag)	n.a.	Partial — results published via Eawag data portals (Eawag)	Eawag data-portal page (Eawag)
Eawag	Sensor Lab (Labor für Sensornetzwerke, Automation und Elektronik) (Eawag)	Swiss surface waters (research infrastructure not a fixed network) (Eawag)	Supports arbitrary parameters; chemical sensing, underwater robotics (Eawag)	n.a. — sites Kastanienbau and Dübendorf (Eawag)	n.a.	n.a.	No — internal research service (Eawag)	Networking and remote access to sensors from signal to data backup (Eawag)
WSL (Swiss Federal Institute for Forest, Snow and Landscape Research)	Hydrometeorological long-term monitoring, Alptal (Erlenbach, Vogelbach, Lümpenbach) (WSL)	Mountain torrents + groundwater (WSL)	Hydrology (runoff), meteorology, snow, water quality, bedload transport, groundwater dynamics, isotope hydrology, nutrient dynamics, morphology (WSL)	3 torrent catchments; snow measured manually at more than 15 locations (WSL)	10-minute means since 1984; daily means 1968–2017; snow weekly to monthly; NADUF water quality weekly since 2003 (WSL)	Direct download of daily means from the WSL data portal EnviDat; sub-daily data on request	Partial — daily means openly downloadable via EnviDat; sub-daily and specialised data only on request;	envidat.ch dataset downloads + researcher requests (WSL)
WSL	EnviDat environmental data portal (WSL)	All environmental domains incl. hydrology/gr datasets (EnviDat)	Dataset-dependent (e.g. experimental hydrological and hydrogeological data of a landslide-prone hillslope) (EnviDat TRAMM)	718 datasets in the catalogue (EnviDat)	Dataset-dependent; near-real-time API for operational stations in the GC-Net case (EnviDat)	CKAN-based; DOI-referenc archives; API endpoints (EnviDat)	Yes — WSL open environmental data repository (WSL); per-dataset licences, name n.a.	CKAN API ('envidat.ch/data-api/...'), DOI downloads, GitHub mirrors (EnviDat)
Existenz.ch (community project, unofficial)	BAFU Hydrology API + SwissMetNet API + custom sensors (Existenz)	Rivers (surface water) + weather (Existenz)	Surface-water temperature, water flow, water level; air temperature, humidity, precipitation, wind, pressure, sunshine, radiation (Existenz)	"Many measurement stations and rivers in Switzerland"; exact count not stated (Existenz)	10-minute source updates; polling every 5 minutes recommended (Existenz)	JSON always; CSV export and metadata API via Datasette; InfluxDB 2 archive with Flux queries; RSS for	Partial — free for public and non-commercial use, unofficial; original owner restrictions (MeteoSwiss,	REST JSON endpoints with CORS '*', optional '&app=/'&version=' parameters, HTTP without SSL possible; Datasette instance and InfluxDB long-term archive (Existenz); open-source parsers 'parse-hydrodaten' / 'parse-swissmetnet' (GitHub)

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opendata.swiss (federal OGD portal)	Water datasets catalogue (BAFU, swisstopo, cantons, EnviDat) (opendata.swiss)	All water body types (metadata catalogue) (opendata.sw)	Catalogue of datasets: water temperature stations, discharge/lake level situation, HUG gauging stations, wastewater monitoring, cantonal hydrometry (opendata.swiss), (opendata.swiss)	n.a. (catalogue)	Per dataset; e.g. water-temperature station dataset updated "kontinuierlich" (opendata.swiss)	Metadata as API (JSON) and XML; resource formats per dataset (opendata.sw	Yes — Swiss OGD portal; individual dataset pages fetched did not name licences (opendata.sw	DCAT metadata API (JSON) / XML download per dataset; filters by organisation and format (opendata.swiss)
Wasserversorgung Zürich (WVZ), Stadt Zürich	WVZ Qualitätsüberwachung (drinking water: lake + groundwater works and distribution network) (Stadt Zürich)	Lake water (Zürichsee), groundwater (Hardhof), treated drinking water and distribution network	Turbidity, pH, conductivity monitored in real time by analysers; microbiological and chemical parameters; open dataset covers more than 300 measurement parameters (biological, metals, pesticides, pharmaceuticals) (Stadt Zürich), (Open Data Zürich)	Station count n.a.; the open dataset lists 9 sampling-poi codes at Seewasserwe Moos, Seewasserwe Lengg and	Continuous/real-time online monitoring at critical points; more than 10,000 lab samples per year; open dataset updated annually (from 2023) (Stadt Zürich), (Open Data Zürich), (Open Data Zürich)	CSV per year ('2023_Trinkw '2024_Trinkw aggregated PDF summary on the WVZ website (Open Data Zürich)	Yes — raw data published machine-read in the Stadt Zürich open-data catalogue (Stadt	data.stadt-zuerich.ch dataset with CSV resources (Open Data Zürich)
SVGW/SSIGE (Schweizerischer Verein des Gas- und Wasserfaches)	trinkwasser.ch quality-data platform (trinkwasser.ch)	Drinking water from public suppliers (trinkwasser.	Supplier-published quality data; parameters not specified on the platform (trinkwasser.ch)	n.a. — one entry per participating water supplier (trinkwasser.	n.a. — explicitly not real time (trinkwasser.ch)	Web pages per supplier; no formats stated (trinkwasser.	No API/licence stated; free publication platform for suppliers (trinkwasser.	Search by postcode / map; supplier login for publication (trinkwasser.ch)

Agency Notes

BAFU / FOEN (Bundesamt für Umwelt).

FOEN is the backbone of Swiss water monitoring: it "operates and coordinates several water monitoring networks" combining long-term fixed stations with single measurements at temporary sites (BAFU). The quantity side (Basismessnetz, ~260 stations, 200 discharge sites) is highly automated — 90 % with automatic remote retrieval — and calibrated by 5–6 discharge measurements per station per year plus weekly on-site checks by local observers (BAFU), (geocat). A real gap: FOEN publishes station geodata openly ("Opendata OPEN: Freie Nutzung") and real-time values via LINDAS at 10-minute cadence, but the measurement-value pages themselves do not state a licence, and the water-quality programmes (NAWA, NAQUA) publish reports and station lists rather than machine-readable measurement series (hydrodaten), (BAFU). Note also two 2026-relevant operational facts: the SMS alerting service ended on 1 July 2026 and users are directed to the "Meine Pegel" push notifications (hydrodaten), and NADUF was renamed NAWA FRACHT in 2025 (BAFU).

NAWA.

NAWA is explicitly limited to flowing waters and does not replace cantonal surveys (BAFU). It is a three-part construct: TREND (~140 sites, uniform parameter set, chem-phys since 2011, biology since 2012, micropollutants since 2018), FRACHT/NADUF (continuous electrode sensors plus 14-day composite chemistry, run jointly with Eawag and WSL since 1972/2003), and SPEZ (campaign monitoring: pesticides 2012/2015/2017/2023, small-stream biology 2018) (BAFU EN). For data interoperability, FOEN maintains a published parameter-name list for chemical substances (version 11/2025) — the reference to use when harmonising cantonal and federal quality data (BAFU).

NAQUA.

More than 600 groundwater monitoring sites in four modules (BAFU). QUANT is the only sensor-heavy module (91 sites, half federal / half cantonal or utility-run, continuous temperature/conductivity/partly turbidity, mostly telemetered; a few cantonal sites are still read out manually, causing delayed data) (BAFU). Note the discrepancy worth flagging in any inventory: the geodata record states "around 100 monitoring sites" while the programme page states 91 (geocat). TREND (50 sites, since 1997/2005) and SPEZ (~500 sites, with the cantons since 2002) are laboratory programmes; ISOT (22 sites) is isotope sampling co-located with MeteoSwiss and NADUF sites (BAFU), (BAFU).

MeteoSwiss.

MeteoSwiss states plainly that monitoring lakes and rivers is not in its remit — discharge, river temperature and lake levels are FOEN's job (MeteoSwiss blog). Its relevance to water monitoring is as precipitation/meteorological input (SwissMetNet ~160 stations + ~100 precipitation stations, 10-minute cadence, fully documented OGD) and as host sites for NAQUA ISOT precipitation sampling (MeteoSwiss), (BAFU).

swisstopo.

No water sensors. swisstopo supplies the hydrographic geometry (swissTLM3D Gewässernetz: watercourses and lake outlines for Switzerland and Liechtenstein, annual updates) and the geo.admin.ch service infrastructure (STAC, ESRI REST, WMS/WMTS) through which FOEN station layers are actually served (opendata.swiss), (geocat).

AWEL Canton Zürich.

Four distinct activities: hydrometry (>60 level stations, near-real-time, discharge derived from rating curves), surface-water quality (15 continuously auto-sampled main stations + 150 periodic sites + 11 lakes + ~100 groundwater/spring abstractions on a 4-year rotation), groundwater levels (continuous data loggers), and groundwater quality (nitrate, conductivity, temperature, oxygen, pH, ions, PFAS, chlorothalonil metabolites since 2019) (Kanton Zürich), (Kanton Zürich), (Kanton Zürich). Access is the weak point: hydrometric data are declared OGD but must be ordered via a form (free for normal requests, CHF 140.20/h for very large ones), with attribution to "AWEL, Kanton Zürich" requested and no API offered; AWEL also refuses to re-deliver BAUFU/MeteoSwiss/other-canton data (Kanton Zürich). Quality work is ISO 17025 accredited and coordinated through the Lab'Eaux network of cantonal water-protection laboratories (Kanton Zürich).

AWA Canton Bern.

The best-quantified cantonal network: ~50 discharge stations, 90 groundwater stations, 120 level gauges and 15 precipitation stations, publishing hydrographic bulletins/yearbooks and a monthly regional groundwater evaluation; data are published online together with federal data but series are obtained through the "Datenbezug" channel and no API or licence is documented (Kanton Bern), (Kanton Bern). The cantonal pages explicitly co-list the federal surface-water and groundwater networks, i.e. federal NAQUA sites in Bern (Niederbipp, Utzenstorf, Worben) are reported through hydrodaten.admin.ch (hydrodaten).

DGE Canton de Vaud (and "SESA").

Vaud's river-quality programme covers 204 sites on more than 50 rivers, running since the 1990s, with biological (macroinvertebrate) and chemical (nutrients, micropollutants) assessment; since 2024 the results are public on the vhv-qualite.ch platform, complementing the "Géoportail de la veille hydrologique vaudoise" for quantity data (info.vd.ch), (Etat de Vaud). Terminology note: "SESA" (Service des eaux, sols et assainissement) is a Vaud service still listed under Canton de Vaud in FOEN's cantonal link list, not a Ticino agency (BAFU). The vhv-qualite.ch platform itself timed out when fetched on 31.08.2026, so its parameters, formats and API could not be confirmed.

Canton Ticino.

Two separate agencies: the Ufficio dei corsi d'acqua runs the quantity network (24 hydrometric + 31 meteorological stations, yearbook + quarterly bulletin, integrated with FOEN and MeteoSwiss networks), and its data are served through the OASI environmental observatory portal, which exposes 39 river stations with discharge and temperature at half-hourly/daily/monthly/annual granularity and an advanced selection-and-download function (Ticino), (OASI). The Ufficio protezione acque e approvvigionamento idrico runs the quality/ecomorphology side using the federal Modular Stepwise Procedure: 986 watercourses (17 % of the ~5,800 km network) analysed, campaigns 2003–2005 and 2021–2022, with critical nitrogen levels downstream of WWTP outfalls in the Vedeggio, Laveggio and Breggia (Ticino).

Other cantons with confirmed sensor networks.

Geneva: 37 stations (12 rain gauges + 25 limnimetric), real-time portal with automatic historical extraction, annual validation lag (Etat de Genève). Aargau: the most transparent sensor documentation of any canton — redundant pressure + radar level probes, water-temperature sensing, PLC cabinets with dual mobile/leased-line transmission, 10-minute data free for public use (Kanton Aargau), plus >50 online groundwater abstraction stations at daily resolution and a joint precipitation network with Luzern (Kanton Aargau). Luzern: 14 river sites, ~40 groundwater/spring sites, records back to 1970, geodata in GDB/SHP/GPKG, but data delivery is form-based and chargeable, with an explicit disclaimer that series may not represent natural conditions precisely (geocat), (Kanton Luzern). Thurgau is the most API-friendly canton: hydrometry stations, water level and water temperature as separate OGD datasets on data.tg.ch with zipped-CSV bulk download and a JSON schema, joined via `obj\_id` — with the caveat that published values are unverified raw sensor data (data.tg.ch), (data.tg.ch). Basel-Stadt publishes continuous Rhine and Birs measurements on data.bs.ch at 15-minute raw resolution with annually plausibilised hourly values (data.bs.ch). FOEN's cantonal link list is the authoritative starting point for the remaining cantons (FR, GL, GR, JU, NE, NW, OW, SH, SO, SG, SZ, UR, ZG, BL), each of which runs its own hydrology or water-protection data service (BAFU).

Eawag.

Eawag is the sensor-innovation hub. The Urban Water Observatory in Fehraltorf is the densest documented water sensor network in Switzerland: 124 sensors in the published dataset (89 on LoRa, 3 LoRaWAN gateways, LORIoT middleware, ~4 km range), 1–5 minute resolution, spanning sewers, the Luppmen/Kempt rivers and groundwater, with an Eawag "Datapool" warehouse, data-access packages for Julia/Python/Matlab and an R-Shiny dashboard (ESSD), (Co-UDlabs). Note the count discrepancy (124 in the dataset vs 111 "currently in the network") — both are cited above. For lakes, Datalakes is the open sensor-to-front-end platform (map viewer, data portal, API, FAIR principles) carrying LéXPLORE and Greifensee high-frequency series, though the API was returning connectivity errors on 31.08.2026 (Datalakes), (Datalakes data portal). Eawag also runs the national wastewater monitoring analytics for BAG (10 WWTPs covering 1.97 M people since January 2025; SARS-CoV-2, influenza A/B, RSV; variant monitoring at 6 WWTPs, published on covSPECTRUM since 2026) (BAG), plus DroMedArio for drug/pharmaceutical consumption and the Groundwater Assessment Platform for global groundwater hazard maps (Eawag). The Sensor

Lab (Dübendorf since 2015, plus Kastanienbaum) provides the chemical-sensing, underwater-robotics and sensor-network engineering behind all of this (Eawag).

WSL.

WSL's water sensing is concentrated in the Alptal torrent research site (Erlenbach, Vogelbach, Lümpenbach): 10-minute means since 1984, daily means 1968–2017 downloadable from EnviDat, weekly NADUF water quality since 2003, snow at >15 manual locations, plus bedload, isotope (ETH Zurich) and catchment groundwater (University of Zurich) data available only on request (WSL). WSL also operates EnviDat, the CKAN-based environmental data repository (718 datasets, DOI-referenced archives, per-dataset API endpoints, harvested into opendata.swiss) (EnviDat), (EnviDat), (opendata.swiss). WSL is also a NAWA FRACHT partner alongside BAFU and Eawag (BAFU).

EPFL / ETH Zurich.

No independent public water sensor networks were confirmed; their monitoring appears through joint infrastructures: LÉXPLORE on Lake Geneva (Eawag + EPFL + UNIL + UNIGE, 10-year deployment, 110 m depth, live skin-temperature feed via Datalakes) (Eawag), the ETH-co-founded Urban Water Observatory with ETH Scientific IT services supporting the Datapool (ESSD), and ETH Zurich's isotope-hydrology instrumentation in WSL's Alptal catchments (WSL).

Aggregators and third-party APIs.

For programmatic access to Swiss hydrology, api.existenz.ch is the practical entry point: JSON-only endpoints wrapping FOEN hydrology and MeteoSwiss SwissMetNet at 10-minute cadence, CORS `\*`, a Datasette instance with CSV export and station/parameter tables, and an InfluxDB 2 long-term archive queried with Flux. It is explicitly unofficial and free for public/non-commercial use, with the original owners' terms applying and BAFU data requiring credit and a link (Existenz), (Datasette). opendata.swiss is a metadata catalogue: DCAT records with API (JSON)/XML metadata access pointing to hydrodaten.admin.ch landing pages; the FOEN dataset pages fetched carried no explicit licence statement, whereas the geocat.ch record for the same station network does ("Opendata OPEN: Freie Nutzung") — so licence status must be checked per catalogue, not per dataset name (opendata.swiss), (geocat).

Drinking water and fountains — an evidence gap.

No national or cantonal network of *fountain* (Brunnen) sensors could be confirmed from any fetched source. The closest confirmed equivalents are: Wasserversorgung Zürich's real-time online analyser network across treatment stages and the distribution network (turbidity, pH, conductivity, with Daphnia biotest units at Grundwasserwerk Hardhof as a last barrier), whose lab results are published as annual CSV files covering >300 parameters (Stadt Zürich), (Open Data Zürich); and SVGW's trinkwasser.ch, a voluntary publication platform for water suppliers that explicitly provides no real-time data and no contamination alerts (trinkwasser.ch). Any project needing fountain-level sensing would have to build it — the market context (LoRaWAN level and quality monitoring for municipalities) exists in Switzerland but was found only in vendor material, not in agency networks.

Known Gaps & Caveats

- Licences are rarely named. Only one fetched page named a licence explicitly: "Opendata OPEN: Freie Nutzung" for the FOEN surface-water station geodata (geocat). Everywhere else "open data" status was asserted without a licence name — including opendata.swiss dataset pages, AWEL's OGD declaration, Aargau's "freier Gebrauch" and Thurgau's OGD portal.
- Sensor brands are essentially never published. Only Aargau (pressure probe + radar probe), FOEN's temperature network (probe + transmitter + logger), NAWA FRACHT (measuring electrodes), Zürich groundwater (modern data loggers) and Eawag/UWO (LoRa/LoRaWAN nodes, LORIOT) disclose technology; no manufacturer or model was found on any fetched page.
- Station counts conflict between sources for NAQUA QUANT (91 vs ~100) and Eawag UWO (124 vs 111); both figures are cited with their source above.
- Not confirmable in this session: vhw-qualite.ch content (fetch timed out), EnviDat dataset-level licences (robots-blocked pages), NAWA/NAQUA measurement-value download endpoints, and station counts for AWEL Zürich groundwater, Ticino UPAAL, and the majority of the smaller cantons.